

A Level Mathematics 9709 · S1

Probability & Statistics 1

全套中英文详细学习手册

Detailed bilingual student manual · concept, derivation, worked examples
and exam method

本册共 8 个独立课时。每课按四页展开：前两页依次讲解 8 个编号知识点；后两页完成 9 项编号化逐题精讲、答案、纠错与评价。题目均为依据考纲原创的训练示例，不复制历年试题。

使用范围与版本 | Scope

0580 册依据 Cambridge IGCSE Mathematics 0580 (20252027) 的 Core/Extended 分层；9709 册依据 Cambridge International AS & A Level Mathematics 9709 (20262027)。M2 单独标为拓展/旧体系衔接材料，不作为现行独立考试组件。

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第 1 课 | 数据表示 | Representation of data

Lesson 1 of 8 · 9709 exam-focused bilingual unit

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 核心定义 | Core definition

图表选择取决于数据类型；箱线图强调中位数与四分位，茎叶图保留原始值。

学习时先用自己的话重述定义，再指出它与相邻概念的区别；这样遇到改变表述的题目仍能识别本质。

English explanation. A statistical diagram is an encoding of data. Read what height, area, position and spread represent before making comparisons, and support every comparison with values.

知识点 02/08 | 符号、公式与条件 | Notation, formulae and conditions

原始数据、茎叶图、箱线图、直方图和累计频数图保留的信息不同。选择图形前先判断数据是离散还是连续、是否分组以及组距是否相等。公式前先写变量含义和允许范围，公式后检查所得值是否满足这些条件。

English explanation. State every parameter and condition before calculation; notation is part of the mathematical model. A formula earns meaning only when its symbols and assumptions are made explicit.

知识点 03/08 | 方法为什么成立 | Why the method works

比较两组必须同时谈位置与离散，并联系情境。

不要把计算器按键当作证明：至少保留建立关系、代入和关键化简三层。

English explanation. Explain the mathematical relationship, not only the calculator sequence. A reader should be able to reproduce the result without guessing which operation was used.

知识点 04/08 | 多种表示之间的转换 | Connecting representations

中位数和IQR抗极端值；均值和标准差使用全部数据。比较两组要同时写中心与离散，并把数值解释回情境。

若图形趋势、代数符号和数表不一致，应暂停并检查模型，而不是继续机械计算。

English explanation. Move deliberately between words, algebra, a graph, a table, a diagram or a probability distribution. Agreement between representations is a powerful error check.

第 1 课（知识点续） | 数据表示 | Representation of data

知识点 05/08 | 高频边界与易错点 | Boundaries and misconceptions

直方图柱面积代表频数，柱高是 $\text{frequency density} = \text{frequency} / \text{class width}$ 。组距不等时绝不能直接以频数作高度。Cambridge 常把这一错误设计成选择题干扰项，结构题则会在后续小问放大前一步错误。

English explanation. A valid method can still give an invalid answer if its domain, sign, unit, interval or model assumptions are ignored. Test the final answer in the original statement.

知识点 06/08 | 基础示范与完整复核 | Worked foundation and verification

A组 median 42, IQR 8; B组 median 39, IQR 15, 比较。正确答案: A typical value higher and results more consistent because its median is larger and IQR smaller.

完成后逆向代回原条件，并说明答案的精度、单位、范围或概率意义。

English explanation. Substitute the result back into the original condition and check its size, sign, accuracy and meaning. Keep exact values until the final requested approximation.

知识点 07/08 | 英文推理与答题句型 | English reasoning language

可使用以下骨架组织答案: From the given information...; Using...; Substituting gives...; Therefore...; This is valid because...。其中 because 后必须写具体数学条件。

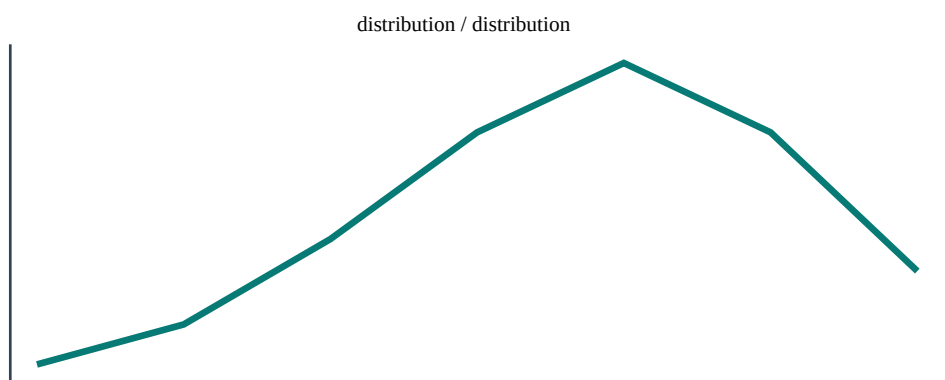
English explanation. A statistical diagram is an encoding of data. Read what height, area, position and spread represent before making comparisons, and support every comparison with values. Use complete sentences to identify the model, justify the operation and interpret the conclusion.

知识点 08/08 | 真题化应用与答案 | Cambridge-style application

Why is a histogram different from a bar chart? 参考答案: Histogram bars touch and area represents frequency for continuous grouped data.

不只核对最终数值，还要对照答案中使用的定义、条件、方法和结论是否齐全。

English explanation. A Cambridge-style response makes the decisive method visible. The correct conclusion here is: Histogram bars touch and area represents frequency for continuous grouped data.



图表/表示检查 | Representation check

图形必须服务于解题：标出坐标、参数、区间、交点或关键方向，并在文字中解释图形与代数/概率表达之间的对应关系。

第 1 课（逐题精讲） | 数据表示 | Representation of data

逐题精讲与评价拓展 | Exam workshop（共 9 项）

精讲 01/09 | 本课真正考什么 | What the lesson examines

题目 | Question. 题目会改变外观，但核心始终是识别对象、条件和目标。先写模型，再计算，最后解释。

Define the object, select a valid method, show decisive working and answer the exact command word.

精讲 02/09 | 示范题 1 | Worked question 1

题目 | Question. A组median 42,IQR 8; B组median 39,IQR 15, 比较。

正确答案 | Correct answer. A typical value higher and results more consistent because its median is larger and IQR smaller. 评分展开:

先写适用公式或关系，再代入题给信息，保留关键中间值，最后给出所需精度与单位。

精讲 03/09 | 选择题 2 | Multiple-choice reasoning

题目 | Question. Which approach is essential for this topic? A Ignore restrictions. B Apply the stated conditions and verify the result. C Round every line. D Give only a calculator display.

正确答案 | Correct answer: B. Conditions determine whether the formula or distribution applies; verification detects invalid roots, events or interpretations.

精讲 04/09 | 结构题 3 | Structured application

题目 | Question. Why is a histogram different from a bar chart?

参考答案 | Model answer. Histogram bars touch and area represents frequency for continuous grouped data. 得分点: 正确识别对象与条件; 选择有效方法; 展示关键运算; 用完整句回答题目。

第 1 课（精讲续） | 数据表示 | Representation of data

精讲 05/09 | 纠错题 4 | Error analysis

题目 | Question. A student writes only the final calculator value and gives no model, formula or event. Explain why the solution may lose marks.

正确答案 | Correct answer. The examiner cannot award method marks or verify that the correct model and event were used. Show the defining formula, substitution and conclusion.

精讲 06/09 | 英文表达 5 | English answer building

题目 | Question. Rewrite the solution as a short mathematical argument.

参考答案 | Model frame. From the given information, define the variables and conditions. Using the relevant result, substitute and simplify. Therefore the required value is ...; it is valid because

精讲 07/09 | 条件变化 6 | Changing a condition

题目 | Question. If one assumption or domain restriction changes, should the original calculation automatically be retained?

正确答案 | Correct answer: No. Recheck the model first. A changed independence condition, interval, parameter, sign or domain can require a different equation or distribution.

精讲 08/09 | 评价拓展 7 | Evaluation and extension

题目 | Question. Evaluate the reliability of the mathematical conclusion.

参考答案 | Model answer. It is reliable only if the stated assumptions hold, the inputs are appropriate, rounding is controlled and the result is interpreted within the original context.

精讲 09/09 | 答案升档 8 | How to improve the response

题目 | Question. What distinguishes a high-quality answer from a bare numerical answer?

正确答案 | Correct answer. It names the method, shows linked steps, preserves exact values where required, reports suitable accuracy and units, includes all valid solutions and states a contextual conclusion.

A Level 大考书写口径 | Cambridge exam focus

Answer structure. Define or identify → state conditions → model → calculate with visible steps → interpret → verify. A calculator value without the distribution, equation, event or decisive method may not receive full credit.

第 2 课 | 排列组合 | Permutations and combinations

Lesson 2 of 8 · 9709 exam-focused bilingual unit

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 核心定义 | Core definition

排列关注顺序，组合只关注选择；限制条件常用分情况或补集。

学习时先用自己的话重述定义，再指出它与相邻概念的区别；这样遇到改变表述的题目仍能识别本质。

English explanation. Counting is a modelling problem. Decide what makes outcomes distinct, handle restrictions explicitly, and use multiplication for successive choices and addition for disjoint cases.

知识点 02/08 | 符号、公式与条件 | Notation, formulae and conditions

排列考虑顺序，组合只考虑选择。阶乘公式是计数工具，不是识别方法；先问交换对象后是否算新结果。

公式前先写变量含义和允许范围，公式后检查所得值是否满足这些条件。

English explanation. State every parameter and condition before calculation; notation is part of the mathematical model. A formula earns meaning only when its symbols and assumptions are made explicit.

知识点 03/08 | 方法为什么成立 | Why the method works

$nPr = \frac{n!}{(n-r)!}$; $nCr = \frac{n!}{r!(n-r)!}$ 。不要把计算器按键当作证明：至少保留建立关系、代入和关键化简三层。

English explanation. Explain the mathematical relationship, not only the calculator sequence. A reader should be able to reproduce the result without guessing which operation was used.

知识点 04/08 | 多种表示之间的转换 | Connecting representations

限制题可用分情况、捆绑、插空或补集。重复字母的排列要除以各重复次数的阶乘。

若图形趋势、代数符号和数表不一致，应暂停并检查模型，而不是继续机械计算。

English explanation. Move deliberately between words, algebra, a graph, a table, a diagram or a probability distribution. Agreement between representations is a powerful error check.

第 2 课（知识点续） | 排列组合 | Permutations and combinations

知识点 05/08 | 高频边界与易错点 | Boundaries and misconceptions

同一题混合选择与排列时分阶段相乘；若不同情况互斥则相加。检查是否重复计数或漏掉合法情况。Cambridge 常把这一错误设计成选择题干扰项，结构题则会在后续小问放大前一步错误。

English explanation. A valid method can still give an invalid answer if its domain, sign, unit, interval or model assumptions are ignored. Test the final answer in the original statement.

知识点 06/08 | 基础示范与完整复核 | Worked foundation and verification

8 人中选 3 人委员会有多少种？正确答案： $8C3=56$ 。
完成后逆向代回原条件，并说明答案的精度、单位、范围或概率意义。

English explanation. Substitute the result back into the original condition and check its size, sign, accuracy and meaning. Keep exact values until the final requested approximation.

知识点 07/08 | 英文推理与答题句型 | English reasoning language

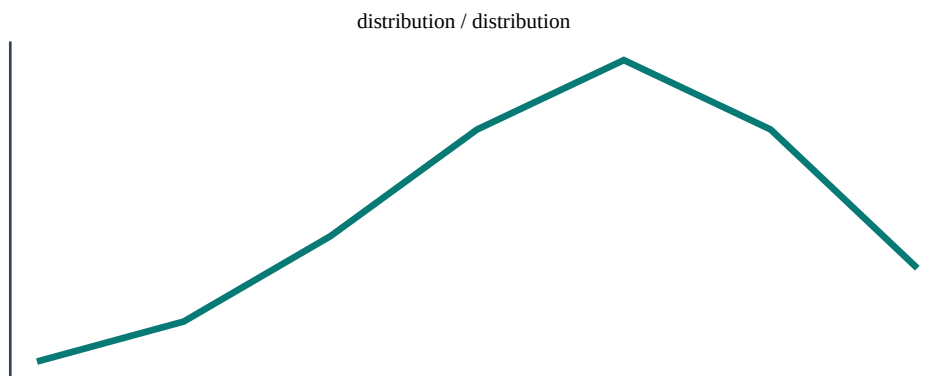
可使用以下骨架组织答案：From the given information...; Using...; Substituting gives...; Therefore...; This is valid because...。其中 because 后必须写具体数学条件。

English explanation. Counting is a modelling problem. Decide what makes outcomes distinct, handle restrictions explicitly, and use multiplication for successive choices and addition for disjoint cases. Use complete sentences to identify the model, justify the operation and interpret the conclusion.

知识点 08/08 | 真题化应用与答案 | Cambridge-style application

How many arrangements of LEVEL? 参考答案： $5!/(2!2!)=30$ because L and E each repeat twice。
不只核对最终数值，还要对照答案中使用的定义、条件、方法和结论是否齐全。

English explanation. A Cambridge-style response makes the decisive method visible. The correct conclusion here is: $5!/(2!2!)=30$ because L and E each repeat twice.



图表/表示检查 | Representation check

图形必须服务于解题：标出坐标、参数、区间、交点或关键方向，并在文字中解释图形与代数/概率表达之间的对应关系。

第 2 课（逐题精讲） | 排列组合 | Permutations and combinations

逐题精讲与评价拓展 | Exam workshop（共 9 项）

精讲 01/09 | 本课真正考什么 | What the lesson examines

题目 | Question. 题目会改变外观，但核心始终是识别对象、条件和目标。先写模型，再计算，最后解释。

Define the object, select a valid method, show decisive working and answer the exact command word.

精讲 02/09 | 示范题 1 | Worked question 1

题目 | Question. 8人中选3人委员会有多少种？

正确答案 | Correct answer. $8C3=56$ 。评分展开：

先写适用公式或关系，再代入题给信息，保留关键中间值，最后给出所需精度与单位。

精讲 03/09 | 选择题 2 | Multiple-choice reasoning

题目 | Question. Which approach is essential for this topic? A Ignore restrictions. B Apply the stated conditions and verify the result. C Round every line. D Give only a calculator display.

正确答案 | Correct answer: B. Conditions determine whether the formula or distribution applies; verification detects invalid roots, events or interpretations.

精讲 04/09 | 结构题 3 | Structured application

题目 | Question. How many arrangements of LEVEL?

参考答案 | Model answer. $5! / (2!2!) = 30$ because L and E each repeat twice. 得分点：

正确识别对象与条件；选择有效方法；展示关键运算；用完整句回答题目。

第 2 课（精讲续） | 排列组合 | Permutations and combinations

精讲 05/09 | 纠错题 4 | Error analysis

题目 | Question. A student writes only the final calculator value and gives no model, formula or event. Explain why the solution may lose marks.

正确答案 | Correct answer. The examiner cannot award method marks or verify that the correct model and event were used. Show the defining formula, substitution and conclusion.

精讲 06/09 | 英文表达 5 | English answer building

题目 | Question. Rewrite the solution as a short mathematical argument.

参考答案 | Model frame. From the given information, define the variables and conditions. Using the relevant result, substitute and simplify. Therefore the required value is ...; it is valid because

精讲 07/09 | 条件变化 6 | Changing a condition

题目 | Question. If one assumption or domain restriction changes, should the original calculation automatically be retained?

正确答案 | Correct answer: No. Recheck the model first. A changed independence condition, interval, parameter, sign or domain can require a different equation or distribution.

精讲 08/09 | 评价拓展 7 | Evaluation and extension

题目 | Question. Evaluate the reliability of the mathematical conclusion.

参考答案 | Model answer. It is reliable only if the stated assumptions hold, the inputs are appropriate, rounding is controlled and the result is interpreted within the original context.

精讲 09/09 | 答案升档 8 | How to improve the response

题目 | Question. What distinguishes a high-quality answer from a bare numerical answer?

正确答案 | Correct answer. It names the method, shows linked steps, preserves exact values where required, reports suitable accuracy and units, includes all valid solutions and states a contextual conclusion.

A Level 大考书写口径 | Cambridge exam focus

Answer structure. Define or identify → state conditions → model → calculate with visible steps → interpret → verify. A calculator value without the distribution, equation, event or decisive method may not receive full credit.

第 3 课 | 概率规则 | Probability rules

Lesson 3 of 8 · 9709 exam-focused bilingual unit

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 核心定义 | Core definition

加法法则处理OR，乘法/条件概率处理AND；独立与互斥绝不能混淆。

学习时先用自己的话重述定义，再指出它与相邻概念的区别；这样遇到改变表述的题目仍能识别本质。

English explanation. Probability notation translates logical words: intersection means “and”, union means inclusive “or”, and conditioning replaces the original sample space with a restricted one.

知识点 02/08 | 符号、公式与条件 | Notation, formulae and conditions

样本空间必须明确。互斥表示不能同时发生；独立表示一个事件不改变另一个概率，两者概念不同。

公式前先写变量含义和允许范围，公式后检查所得值是否满足这些条件。

English explanation. State every parameter and condition before calculation; notation is part of the mathematical model. A formula earns meaning only when its symbols and assumptions are made explicit.

知识点 03/08 | 方法为什么成立 | Why the method works

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ 。不要把计算器按键当作证明：至少保留建立关系、代入和关键化简三层。

English explanation. Explain the mathematical relationship, not only the calculator sequence. A reader should be able to reproduce the result without guessing which operation was used.

知识点 04/08 | 多种表示之间的转换 | Connecting representations

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ ；条件概率 $P(A|B) = P(A \cap B)/P(B)$ 。树图沿枝相乘、互斥路径相加。

若图形趋势、代数符号和数表不一致，应暂停并检查模型，而不是继续机械计算。

English explanation. Move deliberately between words, algebra, a graph, a table, a diagram or a probability distribution. Agreement between representations is a powerful error check.

第 3 课（知识点续） | 概率规则 | Probability rules

知识点 05/08 | 高频边界与易错点 | Boundaries and misconceptions

不放回抽样通常改变后续概率；若声称独立，应验证 $P(A \cap B) = P(A)P(B)$ ，不能凭直觉。Cambridge 常把这一错误设计成选择题干扰项，结构题则会在后续小问放大前一步错误。

English explanation. A valid method can still give an invalid answer if its domain, sign, unit, interval or model assumptions are ignored. Test the final answer in the original statement.

知识点 06/08 | 基础示范与完整复核 | Worked foundation and verification

$P(A)=0.6, P(B)=0.5, P(A \cap B)=0.3$ ，求 $P(A \cup B)$ 。正确答案： $0.6+0.5-0.3=0.8$ 。

完成后逆向代回原条件，并说明答案的精度、单位、范围或概率意义。

English explanation. Substitute the result back into the original condition and check its size, sign, accuracy and meaning. Keep exact values until the final requested approximation.

知识点 07/08 | 英文推理与答题句型 | English reasoning language

可使用以下骨架组织答案：From the given information...; Using...; Substituting gives...; Therefore...; This is valid because...。其中 because 后必须写具体数学条件。

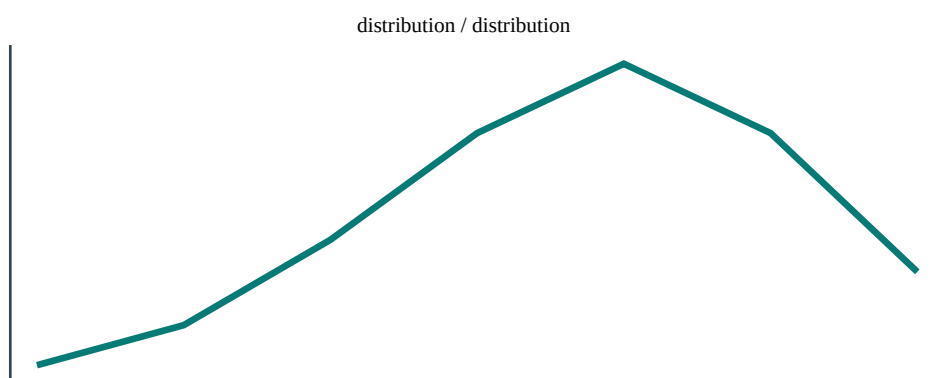
English explanation. Probability notation translates logical words: intersection means “and”, union means inclusive “or”, and conditioning replaces the original sample space with a restricted one. Use complete sentences to identify the model, justify the operation and interpret the conclusion.

知识点 08/08 | 真题化应用与答案 | Cambridge-style application

Are A and B independent? 参考答案：Yes, because $P(A)P(B)=0.6 \times 0.5=0.3=P(A \cap B)$ 。

不只核对最终数值，还要对照答案中使用的定义、条件、方法和结论是否齐全。

English explanation. A Cambridge-style response makes the decisive method visible. The correct conclusion here is: Yes, because $P(A)P(B)=0.6 \times 0.5=0.3=P(A \cap B)$ 。



图表/表示检查 | Representation check

图形必须服务于解题：标出坐标、参数、区间、交点或关键方向，并在文字中解释图形与代数/概率表达之间的对应关系。

第 3 课（逐题精讲） | 概率规则 | Probability rules

逐题精讲与评价拓展 | Exam workshop（共 9 项）

精讲 01/09 | 本课真正考什么 | What the lesson examines

题目 | Question. 题目会改变外观，但核心始终是识别对象、条件和目标。先写模型，再计算，最后解释。

Define the object, select a valid method, show decisive working and answer the exact command word.

精讲 02/09 | 示范题 1 | Worked question 1

题目 | Question. $P(A)=0.6, P(B)=0.5, P(A \cap B)=0.3$, 求 $P(A \cup B)$ 。

正确答案 | Correct answer. $0.6+0.5-0.3=0.8$ 。评分展开:

先写适用公式或关系，再代入题给信息，保留关键中间值，最后给出所需精度与单位。

精讲 03/09 | 选择题 2 | Multiple-choice reasoning

题目 | Question. Which approach is essential for this topic? A Ignore restrictions. B Apply the stated conditions and verify the result. C Round every line. D Give only a calculator display.

正确答案 | Correct answer: B. Conditions determine whether the formula or distribution applies; verification detects invalid roots, events or interpretations.

精讲 04/09 | 结构题 3 | Structured application

题目 | Question. Are A and B independent?

参考答案 | Model answer. Yes, because $P(A)P(B)=0.6 \times 0.5=0.3=P(A \cap B)$. 得分点:

正确识别对象与条件; 选择有效方法; 展示关键运算; 用完整句回答题目。

第 3 课（精讲续） | 概率规则 | Probability rules

精讲 05/09 | 纠错题 4 | Error analysis

题目 | Question. A student writes only the final calculator value and gives no model, formula or event. Explain why the solution may lose marks.

正确答案 | Correct answer. The examiner cannot award method marks or verify that the correct model and event were used. Show the defining formula, substitution and conclusion.

精讲 06/09 | 英文表达 5 | English answer building

题目 | Question. Rewrite the solution as a short mathematical argument.

参考答案 | Model frame. From the given information, define the variables and conditions. Using the relevant result, substitute and simplify. Therefore the required value is ...; it is valid because

精讲 07/09 | 条件变化 6 | Changing a condition

题目 | Question. If one assumption or domain restriction changes, should the original calculation automatically be retained?

正确答案 | Correct answer: No. Recheck the model first. A changed independence condition, interval, parameter, sign or domain can require a different equation or distribution.

精讲 08/09 | 评价拓展 7 | Evaluation and extension

题目 | Question. Evaluate the reliability of the mathematical conclusion.

参考答案 | Model answer. It is reliable only if the stated assumptions hold, the inputs are appropriate, rounding is controlled and the result is interpreted within the original context.

精讲 09/09 | 答案升档 8 | How to improve the response

题目 | Question. What distinguishes a high-quality answer from a bare numerical answer?

正确答案 | Correct answer. It names the method, shows linked steps, preserves exact values where required, reports suitable accuracy and units, includes all valid solutions and states a contextual conclusion.

A Level 大考书写口径 | Cambridge exam focus

Answer structure. Define or identify → state conditions → model → calculate with visible steps → interpret → verify. A calculator value without the distribution, equation, event or decisive method may not receive full credit.

第 4 课 | 离散随机变量 | Discrete random variables

Lesson 4 of 8 · 9709 exam-focused bilingual unit

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 核心定义 | Core definition

概率分布列出X每个可能值及概率；期望是长期平均，不一定是可出现值。
学习时先用自己的话重述定义，再指出它与相邻概念的区别；这样遇到改变表述的题目仍能识别本质。

English explanation. Expectation is a probability-weighted balance point, while variance measures spread in squared units. They summarise a distribution but do not replace the full probability model.

知识点 02/08 | 符号、公式与条件 | Notation, formulae and conditions

随机变量把随机结果映射为数值。分布表必须列出所有可能值，概率非负且总和为1。
公式前先写变量含义和允许范围，公式后检查所得值是否满足这些条件。

English explanation. State every parameter and condition before calculation; notation is part of the mathematical model. A formula earns meaning only when its symbols and assumptions are made explicit.

知识点 03/08 | 方法为什么成立 | Why the method works

$\sum p=1$; $E(X)=\sum xp$; $\text{Var}(X)=E(X)[E(X)]$ 。
不要把计算器按键当作证明：至少保留建立关系、代入和关键化简三层。

English explanation. Explain the mathematical relationship, not only the calculator sequence. A reader should be able to reproduce the result without guessing which operation was used.

知识点 04/08 | 多种表示之间的转换 | Connecting representations

$E(X)=\sum xp$ 是长期平均； $\text{Var}(X)=E(X)-[E(X)]$ 衡量围绕均值的平方离差。
若图形趋势、代数符号和数表不一致，应暂停并检查模型，而不是继续机械计算。

English explanation. Move deliberately between words, algebra, a graph, a table, a diagram or a probability distribution. Agreement between representations is a powerful error check.

第 4 课（知识点续） | 离散随机变量 | Discrete random variables

知识点 05/08 | 高频边界与易错点 | Boundaries and misconceptions

$E(X)$ 不必是 X 可能取到的值；方差不能为负。线性变换满足 $E(aX+b)=aE(X)+b$, $\text{Var}(aX+b)=a^2\text{Var}(X)$ 。

Cambridge 常把这一错误设计成选择题干扰项，结构题则会在后续小问放大前一步错误。

English explanation. A valid method can still give an invalid answer if its domain, sign, unit, interval or model assumptions are ignored. Test the final answer in the original statement.

知识点 06/08 | 基础示范与完整复核 | Worked foundation and verification

X 取 0,1,2，概率 0.2,0.5,0.3，求 $E(X)$ 。正确答案： $0+0.5+0.6=1.1$ 。

完成后逆向代回原条件，并说明答案的精度、单位、范围或概率意义。

English explanation. Substitute the result back into the original condition and check its size, sign, accuracy and meaning. Keep exact values until the final requested approximation.

知识点 07/08 | 英文推理与答题句型 | English reasoning language

可使用以下骨架组织答案：From the given information...; Using...; Substituting gives...; Therefore...; This is valid because...。其中 because 后必须写具体数学条件。

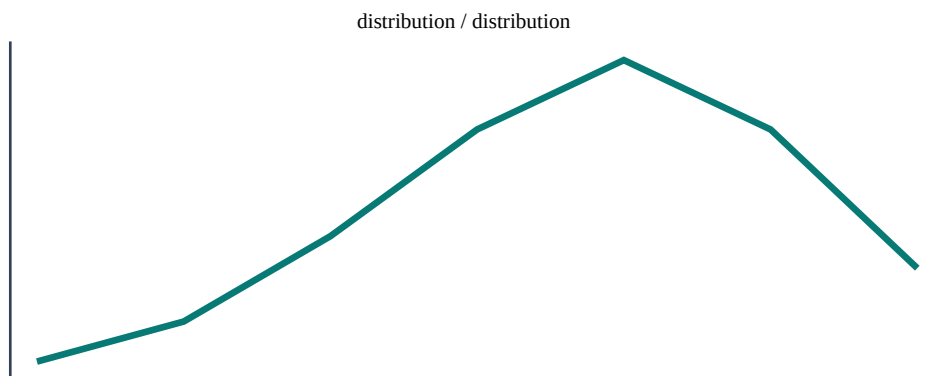
English explanation. Expectation is a probability-weighted balance point, while variance measures spread in squared units. They summarise a distribution but do not replace the full probability model. Use complete sentences to identify the model, justify the operation and interpret the conclusion.

知识点 08/08 | 真题化应用与答案 | Cambridge-style application

Find $\text{Var}(X)$. 参考答案： $E(X)=0+0.5+1.2=1.7$; $\text{variance}=1.7^2-1.1^2=0.49$ 。

不只核对最终数值，还要对照答案中使用的定义、条件、方法和结论是否齐全。

English explanation. A Cambridge-style response makes the decisive method visible. The correct conclusion here is: $E(X)=0+0.5+1.2=1.7$; $\text{variance}=1.7^2-1.1^2=0.49$.



图表/表示检查 | Representation check

图形必须服务于解题：标出坐标、参数、区间、交点或关键方向，并在文字中解释图形与代数/概率表达之间的对应关系。

第 4 课（逐题精讲） | 离散随机变量 | Discrete random variables

逐题精讲与评价拓展 | Exam workshop（共 9 项）

精讲 01/09 | 本课真正考什么 | What the lesson examines

题目 | Question. 题目会改变外观，但核心始终是识别对象、条件和目标。先写模型，再计算，最后解释。

Define the object, select a valid method, show decisive working and answer the exact command word.

精讲 02/09 | 示范题 1 | Worked question 1

题目 | Question. X 取 0, 1, 2, 概率 0.2, 0.5, 0.3, 求 $E(X)$ 。

正确答案 | Correct answer. $0+0.5+0.6=1.1$ 。评分展开:

先写适用公式或关系，再代入题给信息，保留关键中间值，最后给出所需精度与单位。

精讲 03/09 | 选择题 2 | Multiple-choice reasoning

题目 | Question. Which approach is essential for this topic? A Ignore restrictions. B Apply the stated conditions and verify the result. C Round every line. D Give only a calculator display.

正确答案 | Correct answer: B. Conditions determine whether the formula or distribution applies; verification detects invalid roots, events or interpretations.

精讲 04/09 | 结构题 3 | Structured application

题目 | Question. Find $\text{Var}(X)$.

参考答案 | Model answer. $E(X)=0+0.5+1.2=1.7$; $\text{variance}=1.7 \cdot 1=0.49$. 得分点:

正确识别对象与条件; 选择有效方法; 展示关键运算; 用完整句回答题目。

第 4 课（精讲续） | 离散随机变量 | Discrete random variables

精讲 05/09 | 纠错题 4 | Error analysis

题目 | Question. A student writes only the final calculator value and gives no model, formula or event. Explain why the solution may lose marks.

正确答案 | Correct answer. The examiner cannot award method marks or verify that the correct model and event were used. Show the defining formula, substitution and conclusion.

精讲 06/09 | 英文表达 5 | English answer building

题目 | Question. Rewrite the solution as a short mathematical argument.

参考答案 | Model frame. From the given information, define the variables and conditions. Using the relevant result, substitute and simplify. Therefore the required value is ...; it is valid because

精讲 07/09 | 条件变化 6 | Changing a condition

题目 | Question. If one assumption or domain restriction changes, should the original calculation automatically be retained?

正确答案 | Correct answer: No. Recheck the model first. A changed independence condition, interval, parameter, sign or domain can require a different equation or distribution.

精讲 08/09 | 评价拓展 7 | Evaluation and extension

题目 | Question. Evaluate the reliability of the mathematical conclusion.

参考答案 | Model answer. It is reliable only if the stated assumptions hold, the inputs are appropriate, rounding is controlled and the result is interpreted within the original context.

精讲 09/09 | 答案升档 8 | How to improve the response

题目 | Question. What distinguishes a high-quality answer from a bare numerical answer?

正确答案 | Correct answer. It names the method, shows linked steps, preserves exact values where required, reports suitable accuracy and units, includes all valid solutions and states a contextual conclusion.

A Level 大考书写口径 | Cambridge exam focus

Answer structure. Define or identify → state conditions → model → calculate with visible steps → interpret → verify. A calculator value without the distribution, equation, event or decisive method may not receive full credit.

第 5 课 | 二项分布 | Binomial distribution

Lesson 5 of 8 · 9709 exam-focused bilingual unit

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 核心定义 | Core definition

固定次数、两结果、独立、成功概率不变时 $X \sim B(n, p)$ 。

学习时先用自己的话重述定义，再指出它与相邻概念的区别；这样遇到改变表述的题目仍能识别本质。

English explanation. The binomial distribution counts successes in a fixed number of independent Bernoulli trials with constant success probability. Translate words into an event before calculating.

知识点 02/08 | 符号、公式与条件 | Notation, formulae and conditions

$X \sim B(n, p)$ 要求固定试验次数、两类结果、试验独立且成功概率恒定。四条件缺一不可。

公式前先写变量含义和允许范围，公式后检查所得值是否满足这些条件。

English explanation. State every parameter and condition before calculation; notation is part of the mathematical model. A formula earns meaning only when its symbols and assumptions are made explicit.

知识点 03/08 | 方法为什么成立 | Why the method works

$P(X=r) = {}^nC_r p^r (1-p)^{n-r}$; at least常用补集。

不要把计算器按键当作证明：至少保留建立关系、代入和关键化简三层。

English explanation. Explain the mathematical relationship, not only the calculator sequence. A reader should be able to reproduce the result without guessing which operation was used.

知识点 04/08 | 多种表示之间的转换 | Connecting representations

$P(X=r) = {}^nC_r p^r (1-p)^{n-r}$ 。at least、more than和no more than先翻译成不等式，常用补集减少计算。

若图形趋势、代数符号和数表不一致，应暂停并检查模型，而不是继续机械计算。

English explanation. Move deliberately between words, algebra, a graph, a table, a diagram or a probability distribution. Agreement between representations is a powerful error check.

第 5 课（知识点续） | 二项分布 | Binomial distribution

知识点 05/08 | 高频边界与易错点 | Boundaries and misconceptions

识别 n 、 p 和成功事件后再按键。抽样比例过大且不放回时独立性可能不成立。Cambridge 常把这一错误设计成选择题干扰项，结构题则会在后续小问放大前一步错误。

English explanation. A valid method can still give an invalid answer if its domain, sign, unit, interval or model assumptions are ignored. Test the final answer in the original statement.

知识点 06/08 | 基础示范与完整复核 | Worked foundation and verification

$X \sim B(10, 0.2)$ ，求 $P(X=2)$ 。正确答案： $10C2(0.2)(0.8)^8 \approx 0.302$ 。
完成后逆向代回原条件，并说明答案的精度、单位、范围或概率意义。

English explanation. Substitute the result back into the original condition and check its size, sign, accuracy and meaning. Keep exact values until the final requested approximation.

知识点 07/08 | 英文推理与答题句型 | English reasoning language

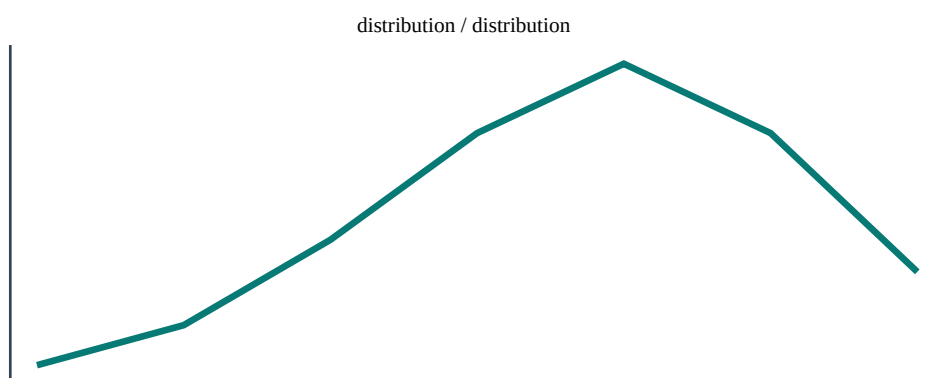
可使用以下骨架组织答案：From the given information...; Using...; Substituting gives...; Therefore...; This is valid because...。其中 because 后必须写具体数学条件。

English explanation. The binomial distribution counts successes in a fixed number of independent Bernoulli trials with constant success probability. Translate words into an event before calculating. Use complete sentences to identify the model, justify the operation and interpret the conclusion.

知识点 08/08 | 真题化应用与答案 | Cambridge-style application

Find $P(X \geq 1)$ 。参考答案： $1 - P(X=0) = 1 - 10.8^{-10} \approx 0.893$ 。
不只核对最终数值，还要对照答案中使用的定义、条件、方法和结论是否齐全。

English explanation. A Cambridge-style response makes the decisive method visible. The correct conclusion here is: $1 - P(X=0) = 1 - 10.8^{-10} \approx 0.893$.



图表/表示检查 | Representation check

图形必须服务于解题：标出坐标、参数、区间、交点或关键方向，并在文字中解释图形与代数/概率表达之间的对应关系。

第 5 课（逐题精讲） | 二项分布 | Binomial distribution

逐题精讲与评价拓展 | Exam workshop（共 9 项）

精讲 01/09 | 本课真正考什么 | What the lesson examines

题目 | Question. 题目会改变外观，但核心始终是识别对象、条件和目标。先写模型，再计算，最后解释。

Define the object, select a valid method, show decisive working and answer the exact command word.

精讲 02/09 | 示范题 1 | Worked question 1

题目 | Question. $X \sim B(10, 0.2)$ ，求 $P(X=2)$ 。

正确答案 | Correct answer. $10C2(0.2)(0.8)^8 \approx 0.302$ 。评分展开：

先写适用公式或关系，再代入题给信息，保留关键中间值，最后给出所需精度与单位。

精讲 03/09 | 选择题 2 | Multiple-choice reasoning

题目 | Question. Which approach is essential for this topic? A Ignore restrictions. B Apply the stated conditions and verify the result. C Round every line. D Give only a calculator display.

正确答案 | Correct answer: B. Conditions determine whether the formula or distribution applies; verification detects invalid roots, events or interpretations.

精讲 04/09 | 结构题 3 | Structured application

题目 | Question. Find $P(X \geq 1)$.

参考答案 | Model answer. $1P(X=0)=10.8^{10} \approx 0.893$. 得分点：

正确识别对象与条件；选择有效方法；展示关键运算；用完整句回答题目。

第 5 课（精讲续） | 二项分布 | Binomial distribution

精讲 05/09 | 纠错题 4 | Error analysis

题目 | Question. A student writes only the final calculator value and gives no model, formula or event. Explain why the solution may lose marks.

正确答案 | Correct answer. The examiner cannot award method marks or verify that the correct model and event were used. Show the defining formula, substitution and conclusion.

精讲 06/09 | 英文表达 5 | English answer building

题目 | Question. Rewrite the solution as a short mathematical argument.

参考答案 | Model frame. From the given information, define the variables and conditions. Using the relevant result, substitute and simplify. Therefore the required value is ...; it is valid because

精讲 07/09 | 条件变化 6 | Changing a condition

题目 | Question. If one assumption or domain restriction changes, should the original calculation automatically be retained?

正确答案 | Correct answer: No. Recheck the model first. A changed independence condition, interval, parameter, sign or domain can require a different equation or distribution.

精讲 08/09 | 评价拓展 7 | Evaluation and extension

题目 | Question. Evaluate the reliability of the mathematical conclusion.

参考答案 | Model answer. It is reliable only if the stated assumptions hold, the inputs are appropriate, rounding is controlled and the result is interpreted within the original context.

精讲 09/09 | 答案升档 8 | How to improve the response

题目 | Question. What distinguishes a high-quality answer from a bare numerical answer?

正确答案 | Correct answer. It names the method, shows linked steps, preserves exact values where required, reports suitable accuracy and units, includes all valid solutions and states a contextual conclusion.

A Level 大考书写口径 | Cambridge exam focus

Answer structure. Define or identify → state conditions → model → calculate with visible steps → interpret → verify. A calculator value without the distribution, equation, event or decisive method may not receive full credit.

第 6 课 | 几何分布 | Geometric distribution

Lesson 6 of 8 · 9709 exam-focused bilingual unit

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 核心定义 | Core definition

等待第一次成功的试验次数服从几何分布。

学习时先用自己的话重述定义，再指出它与相邻概念的区别；这样遇到改变表述的题目仍能识别本质。

English explanation. A geometric variable measures waiting time to the first success. The sequence of failures before the final success explains the power $r-1$.

知识点 02/08 | 符号、公式与条件 | Notation, formulae and conditions

几何分布记录第一次成功出现的试验编号， X 的可能值为 $1, 2, 3, \dots$ 。

公式前先写变量含义和允许范围，公式后检查所得值是否满足这些条件。

English explanation. State every parameter and condition before calculation; notation is part of the mathematical model. A formula earns meaning only when its symbols and assumptions are made explicit.

知识点 03/08 | 方法为什么成立 | Why the method works

$P(X=r)=(1-p)^{r-1}p$; $P(X>r)=(1-p)^r$ 。

不要把计算器按键当作证明：至少保留建立关系、代入和关键化简三层。

English explanation. Explain the mathematical relationship, not only the calculator sequence. A reader should be able to reproduce the result without guessing which operation was used.

知识点 04/08 | 多种表示之间的转换 | Connecting representations

$P(X=r)=(1-p)^{r-1}p$; $P(X>r)=(1-p)^r$ ；无记忆性意味着此前失败不改变未来固定 p 。

若图形趋势、代数符号和数表不一致，应暂停并检查模型，而不是继续机械计算。

English explanation. Move deliberately between words, algebra, a graph, a table, a diagram or a probability distribution. Agreement between representations is a powerful error check.

第 6 课（知识点续） | 几何分布 | Geometric distribution

知识点 05/08 | 高频边界与易错点 | Boundaries and misconceptions

“第 r 次成功”与“ r 次内成功”不同；前者要求前 $r-1$ 次全部失败，第 r 次成功。Cambridge 常把这一错误设计成选择题干扰项，结构题则会在后续小问放大前一步错误。

English explanation. A valid method can still give an invalid answer if its domain, sign, unit, interval or model assumptions are ignored. Test the final answer in the original statement.

知识点 06/08 | 基础示范与完整复核 | Worked foundation and verification

成功率0.3，首次成功在第4次概率。正确答案： $0.7 \times 0.3 = 0.1029$ 。
完成后逆向代回原条件，并说明答案的精度、单位、范围或概率意义。

English explanation. Substitute the result back into the original condition and check its size, sign, accuracy and meaning. Keep exact values until the final requested approximation.

知识点 07/08 | 英文推理与答题句型 | English reasoning language

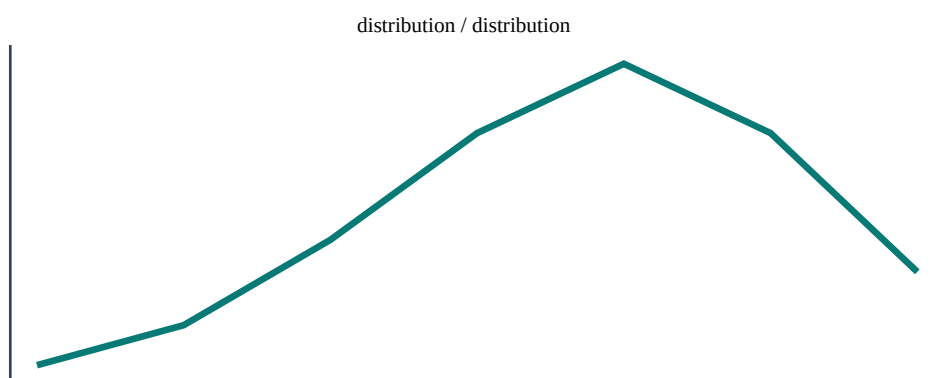
可使用以下骨架组织答案：From the given information...; Using...; Substituting gives...; Therefore...; This is valid because...。其中 because 后必须写具体数学条件。

English explanation. A geometric variable measures waiting time to the first success. The sequence of failures before the final success explains the power $r-1$. Use complete sentences to identify the model, justify the operation and interpret the conclusion.

知识点 08/08 | 真题化应用与答案 | Cambridge-style application

Find probability first success occurs after trial 5. 参考答案： $0.7^5 \approx 0.1681$ 。
不只核对最终数值，还要对照答案中使用的定义、条件、方法和结论是否齐全。

English explanation. A Cambridge-style response makes the decisive method visible. The correct conclusion here is: $0.7^5 \approx 0.1681$.



图表/表示检查 | Representation check

图形必须服务于解题：标出坐标、参数、区间、交点或关键方向，并在文字中解释图形与代数/概率表达之间的对应关系。

第 6 课（逐题精讲） | 几何分布 | Geometric distribution

逐题精讲与评价拓展 | Exam workshop（共 9 项）

精讲 01/09 | 本课真正考什么 | What the lesson examines

题目 | Question. 题目会改变外观，但核心始终是识别对象、条件和目标。先写模型，再计算，最后解释。

Define the object, select a valid method, show decisive working and answer the exact command word.

精讲 02/09 | 示范题 1 | Worked question 1

题目 | Question. 成功率0.3，首次成功在第4次概率。

正确答案 | Correct answer. $0.7 \times 0.3 = 0.1029$ 。评分展开：

先写适用公式或关系，再代入题给信息，保留关键中间值，最后给出所需精度与单位。

精讲 03/09 | 选择题 2 | Multiple-choice reasoning

题目 | Question. Which approach is essential for this topic? A Ignore restrictions. B Apply the stated conditions and verify the result. C Round every line. D Give only a calculator display.

正确答案 | Correct answer: B. Conditions determine whether the formula or distribution applies; verification detects invalid roots, events or interpretations.

精讲 04/09 | 结构题 3 | Structured application

题目 | Question. Find probability first success occurs after trial 5.

参考答案 | Model answer. $0.7^5 \approx 0.1681$. 得分点：

正确识别对象与条件；选择有效方法；展示关键运算；用完整句回答题目。

第 6 课（精讲续） | 几何分布 | Geometric distribution

精讲 05/09 | 纠错题 4 | Error analysis

题目 | Question. A student writes only the final calculator value and gives no model, formula or event. Explain why the solution may lose marks.

正确答案 | Correct answer. The examiner cannot award method marks or verify that the correct model and event were used. Show the defining formula, substitution and conclusion.

精讲 06/09 | 英文表达 5 | English answer building

题目 | Question. Rewrite the solution as a short mathematical argument.

参考答案 | Model frame. From the given information, define the variables and conditions. Using the relevant result, substitute and simplify. Therefore the required value is ...; it is valid because

精讲 07/09 | 条件变化 6 | Changing a condition

题目 | Question. If one assumption or domain restriction changes, should the original calculation automatically be retained?

正确答案 | Correct answer: No. Recheck the model first. A changed independence condition, interval, parameter, sign or domain can require a different equation or distribution.

精讲 08/09 | 评价拓展 7 | Evaluation and extension

题目 | Question. Evaluate the reliability of the mathematical conclusion.

参考答案 | Model answer. It is reliable only if the stated assumptions hold, the inputs are appropriate, rounding is controlled and the result is interpreted within the original context.

精讲 09/09 | 答案升档 8 | How to improve the response

题目 | Question. What distinguishes a high-quality answer from a bare numerical answer?

正确答案 | Correct answer. It names the method, shows linked steps, preserves exact values where required, reports suitable accuracy and units, includes all valid solutions and states a contextual conclusion.

A Level 大考书写口径 | Cambridge exam focus

Answer structure. Define or identify → state conditions → model → calculate with visible steps → interpret → verify. A calculator value without the distribution, equation, event or decisive method may not receive full credit.

第 7 课 | 正态分布 | Normal distribution

Lesson 7 of 8 · 9709 exam-focused bilingual unit

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 核心定义 | Core definition

正态分布由 μ 和 σ 决定，对称且总面积 1；标准化 $Z = (X - \mu) / \sigma$ 。

学习时先用自己的话重述定义，再指出它与相邻概念的区别；这样遇到改变表述的题目仍能识别本质。

English explanation. Normal calculations are area calculations. A sketch fixes the correct tail, standardisation measures distance in standard deviations, and continuity correction aligns discrete bars with continuous area.

知识点 02/08 | 符号、公式与条件 | Notation, formulae and conditions

$X \sim N(\mu, \sigma)$ 中的第二参数是方差，不是标准差。曲线关于 μ 对称，总面积为 1。

公式前先写变量含义和允许范围，公式后检查所得值是否满足这些条件。

English explanation. State every parameter and condition before calculation; notation is part of the mathematical model. A formula earns meaning only when its symbols and assumptions are made explicit.

知识点 03/08 | 方法为什么成立 | Why the method works

画草图、标均值与阴影，再用表/计算器；反求值时从概率找 z 。

不要把计算器按键当作证明：至少保留建立关系、代入和关键化简三层。

English explanation. Explain the mathematical relationship, not only the calculator sequence. A reader should be able to reproduce the result without guessing which operation was used.

知识点 04/08 | 多种表示之间的转换 | Connecting representations

标准化 $Z = (X - \mu) / \sigma$ ；先画阴影、写概率事件，再使用表或计算器。反求界值时从尾部概率确定 z 符号。

若图形趋势、代数符号和数表不一致，应暂停并检查模型，而不是继续机械计算。

English explanation. Move deliberately between words, algebra, a graph, a table, a diagram or a probability distribution. Agreement between representations is a powerful error check.

第 7 课（知识点续） | 正态分布 | Normal distribution

知识点 05/08 | 高频边界与易错点 | Boundaries and misconceptions

连续分布单点概率为0。二项近似正态时检查 np 与 $n(1-p)$ 足够大，并做 ± 0.5 连续性修正。Cambridge常把这一错误设计成选择题干扰项，结构题则会在后续小问放大前一步错误。

English explanation. A valid method can still give an invalid answer if its domain, sign, unit, interval or model assumptions are ignored. Test the final answer in the original statement.

知识点 06/08 | 基础示范与完整复核 | Worked foundation and verification

$X \sim N(50, 9)$ ，求 $P(X < 56)$ 。正确答案： $\sigma = 3$ ， $z = (56 - 50)/3 = 2$ ， $P \approx 0.9772$ 。

完成后逆向代回原条件，并说明答案的精度、单位、范围或概率意义。

English explanation. Substitute the result back into the original condition and check its size, sign, accuracy and meaning. Keep exact values until the final requested approximation.

知识点 07/08 | 英文推理与答题句型 | English reasoning language

可使用以下骨架组织答案：From the given information...; Using...; Substituting gives...; Therefore...; This is valid because...。其中 because 后必须写具体数学条件。

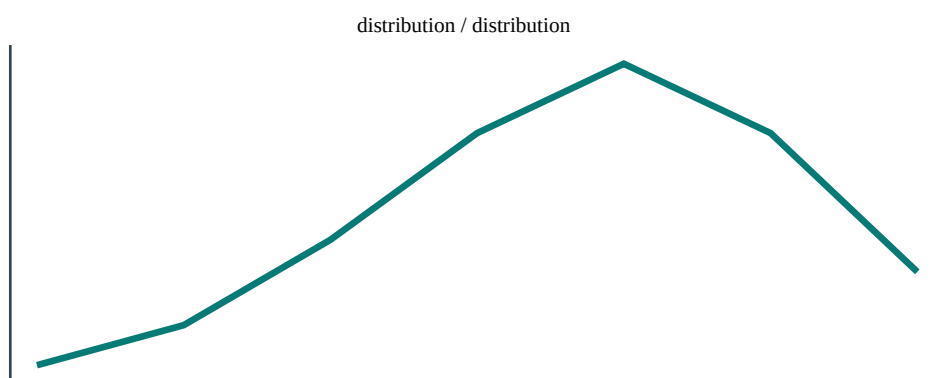
English explanation. Normal calculations are area calculations. A sketch fixes the correct tail, standardisation measures distance in standard deviations, and continuity correction aligns discrete bars with continuous area. Use complete sentences to identify the model, justify the operation and interpret the conclusion.

知识点 08/08 | 真题化应用与答案 | Cambridge-style application

Find a with $P(X < a) = 0.1$ 。参考答案： $z \approx 1.2816$ ，so $a = 50 + 1.2816 \times 3 \approx 46.2$ 。

不只核对最终数值，还要对照答案中使用的定义、条件、方法和结论是否齐全。

English explanation. A Cambridge-style response makes the decisive method visible. The correct conclusion here is: $z \approx 1.2816$ ，so $a = 50 + 1.2816 \times 3 \approx 46.2$ 。



图表/表示检查 | Representation check

图形必须服务于解题：标出坐标、参数、区间、交点或关键方向，并在文字中解释图形与代数/概率表达之间的对应关系。

第 7 课（逐题精讲） | 正态分布 | Normal distribution

逐题精讲与评价拓展 | Exam workshop（共 9 项）

精讲 01/09 | 本课真正考什么 | What the lesson examines

题目 | Question. 题目会改变外观，但核心始终是识别对象、条件和目标。先写模型，再计算，最后解释。

Define the object, select a valid method, show decisive working and answer the exact command word.

精讲 02/09 | 示范题 1 | Worked question 1

题目 | Question. $X \sim N(50, 9)$ ，求 $P(X < 56)$ 。

正确答案 | Correct answer. $\sigma = 3$, $z = (56 - 50)/3 = 2$, $P \approx 0.9772$ 。评分展开：

先写适用公式或关系，再代入题给信息，保留关键中间值，最后给出所需精度与单位。

精讲 03/09 | 选择题 2 | Multiple-choice reasoning

题目 | Question. Which approach is essential for this topic? A Ignore restrictions. B Apply the stated conditions and verify the result. C Round every line. D Give only a calculator display.

正确答案 | Correct answer: B. Conditions determine whether the formula or distribution applies; verification detects invalid roots, events or interpretations.

精讲 04/09 | 结构题 3 | Structured application

题目 | Question. Find a with $P(X < a) = 0.1$.

参考答案 | Model answer. $z \approx 1.2816$, so $a = 50 + 1.2816 \times 3 \approx 53.8$. 得分点：

正确识别对象与条件；选择有效方法；展示关键运算；用完整句回答题目。

第 7 课（精讲续） | 正态分布 | Normal distribution

精讲 05/09 | 纠错题 4 | Error analysis

题目 | Question. A student writes only the final calculator value and gives no model, formula or event. Explain why the solution may lose marks.

正确答案 | Correct answer. The examiner cannot award method marks or verify that the correct model and event were used. Show the defining formula, substitution and conclusion.

精讲 06/09 | 英文表达 5 | English answer building

题目 | Question. Rewrite the solution as a short mathematical argument.

参考答案 | Model frame. From the given information, define the variables and conditions. Using the relevant result, substitute and simplify. Therefore the required value is ...; it is valid because

精讲 07/09 | 条件变化 6 | Changing a condition

题目 | Question. If one assumption or domain restriction changes, should the original calculation automatically be retained?

正确答案 | Correct answer: No. Recheck the model first. A changed independence condition, interval, parameter, sign or domain can require a different equation or distribution.

精讲 08/09 | 评价拓展 7 | Evaluation and extension

题目 | Question. Evaluate the reliability of the mathematical conclusion.

参考答案 | Model answer. It is reliable only if the stated assumptions hold, the inputs are appropriate, rounding is controlled and the result is interpreted within the original context.

精讲 09/09 | 答案升档 8 | How to improve the response

题目 | Question. What distinguishes a high-quality answer from a bare numerical answer?

正确答案 | Correct answer. It names the method, shows linked steps, preserves exact values where required, reports suitable accuracy and units, includes all valid solutions and states a contextual conclusion.

A Level 大考书写口径 | Cambridge exam focus

Answer structure. Define or identify → state conditions → model → calculate with visible steps → interpret → verify. A calculator value without the distribution, equation, event or decisive method may not receive full credit.

第 8 课 | S1综合建模 | Statistics synthesis

Lesson 8 of 8 · 9709 exam-focused bilingual unit

深度展开与真题化应用 | In-depth development (共 8 个知识点)

知识点 01/08 | 核心定义 | Core definition

先识别模型条件，再计算；最终概率必须在0到1之间。

学习时先用自己的话重述定义，再指出它与相邻概念的区别；这样遇到改变表述的题目仍能识别本质。

English explanation. A complete statistics solution names the random variable, justifies the model, states parameters and event, calculates transparently, and interprets the final probability.

知识点 02/08 | 符号、公式与条件 | Notation, formulae and conditions

综合题先判断数据类型、事件结构与分布条件，再选择统计量或概率模型。计算器指令不能替代模型说明。

公式前先写变量含义和允许范围，公式后检查所得值是否满足这些条件。

English explanation. State every parameter and condition before calculation; notation is part of the mathematical model. A formula earns meaning only when its symbols and assumptions are made explicit.

知识点 03/08 | 方法为什么成立 | Why the method works

独立、固定 p 或正态近似的条件需从题干验证。

不要把计算器按键当作证明：至少保留建立关系、代入和关键化简三层。

English explanation. Explain the mathematical relationship, not only the calculator sequence. A reader should be able to reproduce the result without guessing which operation was used.

知识点 04/08 | 多种表示之间的转换 | Connecting representations

作答顺序：定义 X ；写 X 的分布及参数；写目标概率；显示关键代入；给适当精度；解释答案。

若图形趋势、代数符号和数表不一致，应暂停并检查模型，而不是继续机械计算。

English explanation. Move deliberately between words, algebra, a graph, a table, a diagram or a probability distribution. Agreement between representations is a powerful error check.

第 8 课（知识点续） | S1综合建模 | Statistics synthesis

知识点 05/08 | 高频边界与易错点 | Boundaries and misconceptions

结果检查包括 $0 \leq P \leq 1$ 、方差非负、计数为整数，以及近似条件是否满足。Cambridge 常把这一错误设计成选择题干扰项，结构题则会在后续小问放大前一步错误。

English explanation. A valid method can still give an invalid answer if its domain, sign, unit, interval or model assumptions are ignored. Test the final answer in the original statement.

知识点 06/08 | 基础示范与完整复核 | Worked foundation and verification

何时不能用二项模型？正确答案：试验不独立、成功概率改变、次数不固定或结果不止成功/失败时。完成后逆向代回原条件，并说明答案的精度、单位、范围或概率意义。

English explanation. Substitute the result back into the original condition and check its size, sign, accuracy and meaning. Keep exact values until the final requested approximation.

知识点 07/08 | 英文推理与答题句型 | English reasoning language

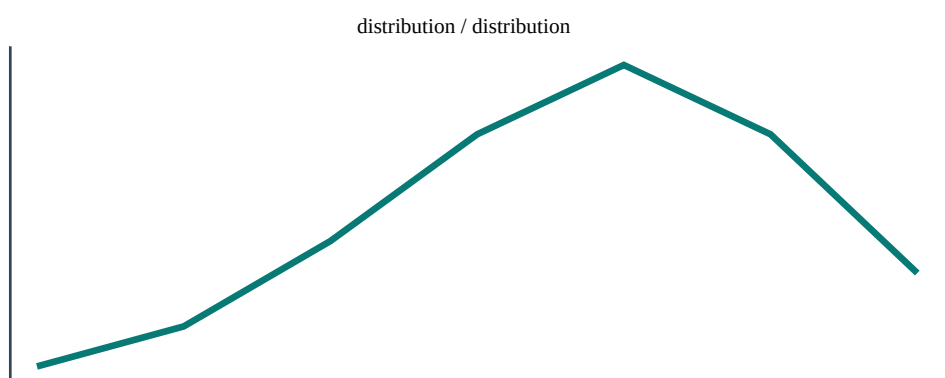
可使用以下骨架组织答案：From the given information...; Using...; Substituting gives...; Therefore...; This is valid because...。其中 because 后必须写具体数学条件。

English explanation. A complete statistics solution names the random variable, justifies the model, states parameters and event, calculates transparently, and interprets the final probability. Use complete sentences to identify the model, justify the operation and interpret the conclusion.

知识点 08/08 | 真题化应用与答案 | Cambridge-style application

Why should calculator notation be translated? 参考答案：An answer needs the named distribution, parameters, event and probability, not only an unexplained decimal.
不只核对最终数值，还要对照答案中使用的定义、条件、方法和结论是否齐全。

English explanation. A Cambridge-style response makes the decisive method visible. The correct conclusion here is: An answer needs the named distribution, parameters, event and probability, not only an unexplained decimal.



图表/表示检查 | Representation check

图形必须服务于解题：标出坐标、参数、区间、交点或关键方向，并在文字中解释图形与代数/概率表达之间的对应关系。

第 8 课（逐题精讲） | S1综合建模 | Statistics synthesis

逐题精讲与评价拓展 | Exam workshop（共 9 项）

精讲 01/09 | 本课真正考什么 | What the lesson examines

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Define the object, select a valid method, show decisive working and answer the exact command word.

精讲 02/09 | 示范题 1 | Worked question 1

题目 | Question. 何时不能用二项模型？

正确答案 | Correct answer. 试验不独立、成功概率改变、次数不固定或结果不止成功/失败时。评分展开：先写适用公式或关系，再代入题给信息，保留关键中间值，最后给出所需精度与单位。

精讲 03/09 | 选择题 2 | Multiple-choice reasoning

题目 | Question. Which approach is essential for this topic? A Ignore restrictions. B Apply the stated conditions and verify the result. C Round every line. D Give only a calculator display.

正确答案 | Correct answer: B. Conditions determine whether the formula or distribution applies; verification detects invalid roots, events or interpretations.

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题目 | Question. Why should calculator notation be translated?

参考答案 | Model answer. An answer needs the named distribution, parameters, event and probability, not only an unexplained decimal. 得分点：

正确识别对象与条件；选择有效方法；展示关键运算；用完整句回答题目。

第 8 课（精讲续） | S1综合建模 | Statistics synthesis

精讲 05/09 | 纠错题 4 | Error analysis

题目 | Question. A student writes only the final calculator value and gives no model, formula or event. Explain why the solution may lose marks.

正确答案 | Correct answer. The examiner cannot award method marks or verify that the correct model and event were used. Show the defining formula, substitution and conclusion.

精讲 06/09 | 英文表达 5 | English answer building

题目 | Question. Rewrite the solution as a short mathematical argument.

参考答案 | Model frame. From the given information, define the variables and conditions. Using the relevant result, substitute and simplify. Therefore the required value is ...; it is valid because

精讲 07/09 | 条件变化 6 | Changing a condition

题目 | Question. If one assumption or domain restriction changes, should the original calculation automatically be retained?

正确答案 | Correct answer: No. Recheck the model first. A changed independence condition, interval, parameter, sign or domain can require a different equation or distribution.

精讲 08/09 | 评价拓展 7 | Evaluation and extension

题目 | Question. Evaluate the reliability of the mathematical conclusion.

参考答案 | Model answer. It is reliable only if the stated assumptions hold, the inputs are appropriate, rounding is controlled and the result is interpreted within the original context.

精讲 09/09 | 答案升档 8 | How to improve the response

题目 | Question. What distinguishes a high-quality answer from a bare numerical answer?

正确答案 | Correct answer. It names the method, shows linked steps, preserves exact values where required, reports suitable accuracy and units, includes all valid solutions and states a contextual conclusion.

A Level 大考书写口径 | Cambridge exam focus

Answer structure. Define or identify → state conditions → model → calculate with visible steps → interpret → verify. A calculator value without the distribution, equation, event or decisive method may not receive full credit.